TURKIC INTERNATIONAL MATHEMATICAL OLYMPIAD

TIMO

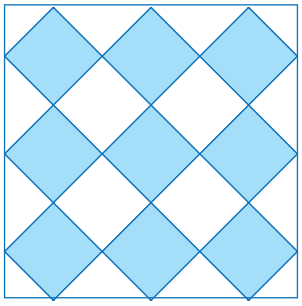
GRADE 6

Mathematical Horizons: Uniting Turkic Minds, Expanding Global Genius.

SCHOLARSHIP FROM TURKISH UNIVERSITIES

MEDIPOL · BAHÇEŞEHİR · İSTANBUL · İSTANBUL TICARET · İSTANBUL GALATA ÜNİVERSİTESİ

1. All of the small squares given in the figure below are in the same size.Все маленькие



According to this, the sum of the areas of the coloured small squares is what fraction of the area of the large square?

A) 1 B) $\frac{1}{2}$ C) $\frac{1}{3}$ D) $\frac{2}{3}$ E) $\frac{3}{4}$

2. Elif, İnci, Kayra and Zeynep are four siblings. The following information is given about the ages of these siblings, whose ages are 8, 13, 17 and 19.

- Elif is younger than Kayra.
- The sum of the ages of Elif and Zeynep is divisible by 5.
- The sum of the ages of Kayra and Zeynep is divisible by 5.

According to this what is the difference of the ages of İnci and Elif?

A) 4 B) 5 C) 6 D) 9 E) 11

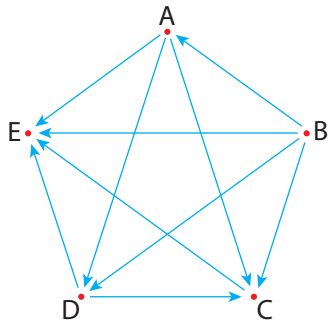
3. Maya will choose one geometric shape from each of the following groups.

Group 1: △, ☆, □, ⬡, ▽
Group 2: ☆, ⬢, ⬟, ⊓
Group 3: □, ⬢, ⬢

How many different ways can Maya make this choice, provided that she has 3 different geometric patternseach time?

A) 42 B) 44 C) 46 D) 48 E) 37

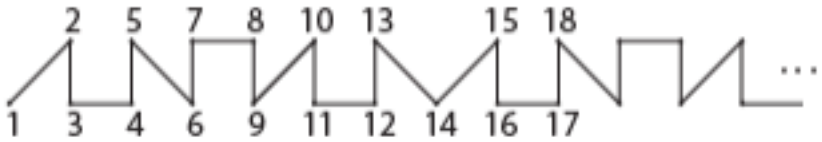
4. The figure below shows the relationships of the ages of the cousins labelled with the letters A, B, C, D and E with arrows.
A → B means the cousin A is older than the cousin B.



Considering this diagram, if one orders the cousins from younger to older, which one will be the 2nd person from the beginning?

A) A B) B C) C D) D E) E

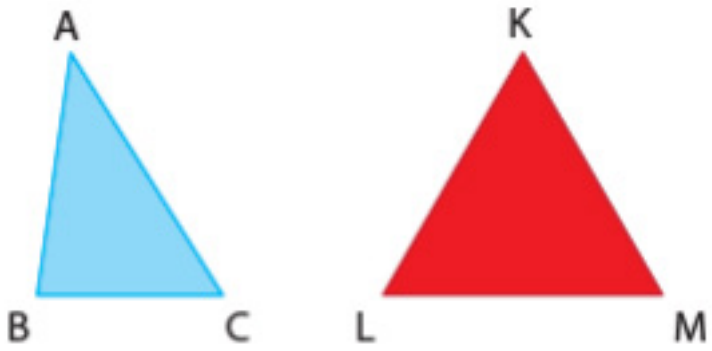
5. The diagram below numbered points are connected by the line segments. In addition, every 14 points repeat themselves in this diagram.



According to this, which of the following is the diagram that includes the points from 2022 to 2025?

A)
B)
C)
D)
E)

6.



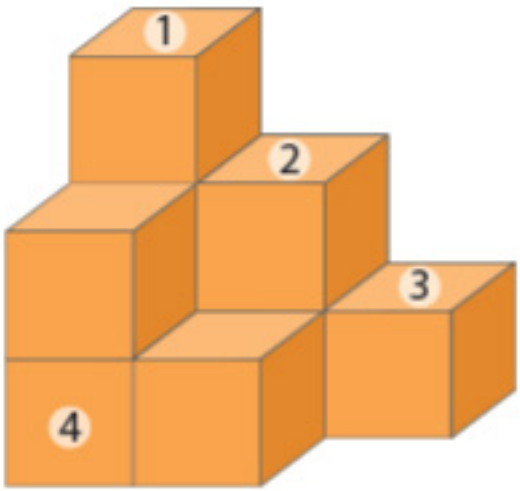
The circumference of the blue triangle given above is 29 cm and the circumference of the red, equileteral, triangle is 42 cm. It is known that one of the length of the side of the blue triangle and the length of the side of the red triangle is equal to each other.
What is the perimeter of the quadrilateral obtained by matching these equal sides?

A) 50 B) 43 C) 41 D) 39 E) 36

7. It is known that the number of spectators who come to watch the basketball team play in a school is greater than 500 and less than 700. It is also known that the number of spectators is an even number with different digits.
According to this, how many different values can the number of spectators take?

A) 36 B) 42 C) 56 D) 72 E) 92

8. Using 9 identical cubes, the following figure is obtained.



Taking which of the numbered cubes will not change the surface area of the figure?

A) 1,2 B) 2,3 C) 3 D) 4 E) 2

9. Ahmet has 86 blue marbles, 52 yellow marbles and 106 green marbles in his bag. Ahmet wants to distribute all of these marbles to each of his friends by giving one marble of 2 different colours to each of them.
According to this, how many friends of Ahmet have yellow and green marbles?

A) 46 B) 36 C) 28 D) 24 E) 22

10. When identical glasses are nested, the distance between the tops of the two glasses is always the same.

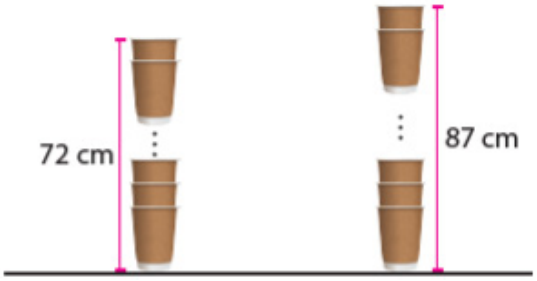


Figure-1 Figure-2

The height of 21 glasses in Figure-1 is 72 cm and the height of 26 glasses in Figure-2 is 87 cm.
According to this, how many cm is the height of 100 nested glasses?

A) 294 B) 297 C) 300 D) 306 E) 309

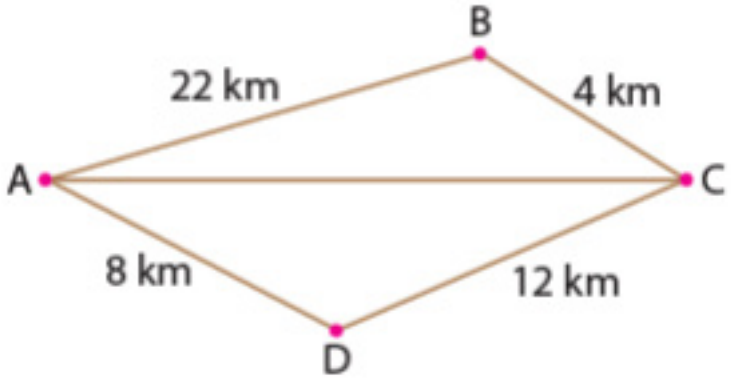
11. Five books that Onur wants to buy and their prices are shown in the table below.

Book Name	Price
Perili Köşk	70 ₺
Guliver'in Gezileri	90 ₺
Pinokyo	80 ₺
Küçük Prens	60 ₺
Nasreddin Hoca	75 ₺

Onur's money is not enough to buy any four of the books, but it is enough to buy any three of them.
According to this, how many different natural number values can Onur's money be in terms of ₺?

A) 39 B) 40 C) 41 D) 42 E) 43

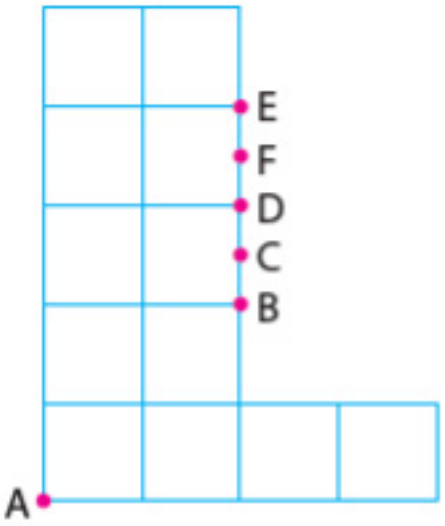
12. “In a triangle, the sum of the lengths of two sides is greater than the third side.”
Below is a sketch of 4 villages and the distances between the villages. It is known that the distances between these villages are integers.



According to this, what is the distance between the village A and the village C?

A) 17 B) 18 C) 19 D) 20 E) 21

13.



$|BC| = |CD| = |DF|$

The figure above consists of unit squares.
If which of the points B, C, D and E is connected with the point A, the figure will be divided into two parts with equal areas?

A) B B) C C) D D) E E) F

14. In a class, boys shook hands with each other and girls shook hands with each other once. The number of handshakes of boys is 15 more than the number of handshakes of girls.
Since the number of boys is 3 more than girls in this class, howmany students are there in this class?

A) 10 B) 11 C) 12 D) 13 E) 15

15.

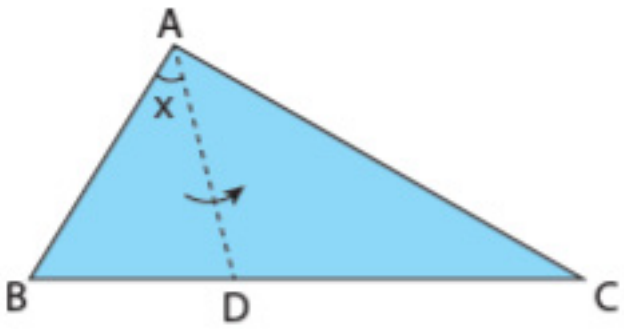


Figure-1

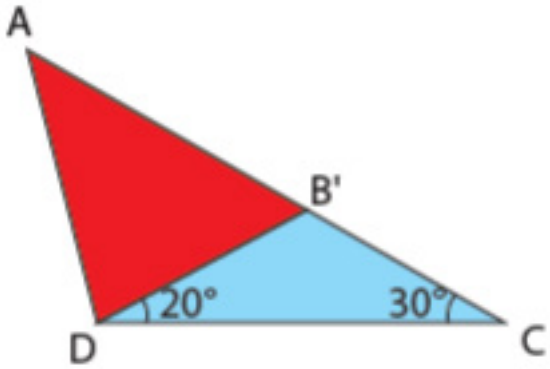


Figure-2

When the triangle ABC given in Figure-1 is folded along AD by holding the corner B, the corner B comes to the point B' on [AC].

Since $m(B'\widehat{D}C) = 20°$ and $m(B'\widehat{C}D) = 30°$, what is the measure of the angle $m(B\widehat{A}D)$?

A) 30 B) 40 C) 50 D) 65 E) 75

16. Natural numbers from 1 to 16 are written as follows.

1	2	3	4
8	7	6	5
9	10	11	12
16	15	14	13

Select 4 of these numbers, any two of which are not in the same row or in the same column.
What is the greatest possible value of the sum of these selected 4 numbers?

A) 39 B) 38 C) 37 D) 36 E) 42

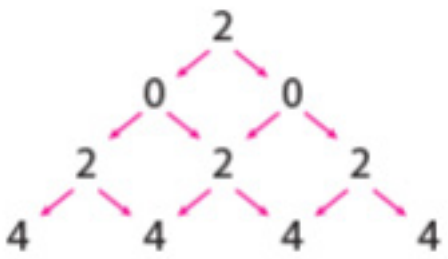
17. The numbers written in the table below are obtained from left to right and from top to bottom,

1	?		
3			
5		15	
7			22

According to this what is the number marked by the question mark?

A) 7 B) 6 C) 5 D) 4 E) 3

18. In the figure below, starting with the number 2 and ending with one of the numbers 4 in the bottom row, the number 2024 will be formed.



According by following the arrows, howmany “2024” can be obtained?

A) 4 B) 6 C) 8 D) 10 E) 12

19. An operation is defined with the numbers in the figure as folows

a	b
c	d

$\rightarrow a \times d + b \times c$

1	2
3	4

the operation gives the result 1X4+2X3=10.

3	5
7	?

If for the figure the result of the operation is 68 what is the number denoted by the question mark?

A) 9 B) 10 C) 11 D) 12 E) 14

20. A game is called fair if the chances of any player's winning and losing are equal to each other. According to this, which of the following dice throwing games is a fair game? (The numbers 1, 2, 3, 4, 5 and 6 are written on the surfaces of the dice.)

A) Throwing 3 wins, the others looses.
B) Throwing a prime number wins, the others looses.
C) Throwing a number greater than 4 wins, the others looses.
D) Throwing a number divisible by 3 wins, the others looses.
E) Throwing a number less than 3 wins, the others looses.

21. The length of a side of a cube is 1 unit long. An ant walks along the sides of this cube without passing through the same partition twice.
According to this, what is the length of the longest path that the ant can walk? (The ant can pass through the same corner several times.)

A) 8 B) 9 C) 10 D) 11 E) 12

22. In a class of 30 students, 3 different books will be distributed to each student. These books belong to maths, physics, chemistry and biology courses. The books of a course are identical in themselves .
Since a total of 25, 24 and 21 maths, physics and chemistry books were distributed respectively, how many students did not receive a biology book?

A) 11 B) 10 C) 9 D) 8 E) None of them

23. If we delete 3 in the number 20312, we get the number 2012.
According to this, how many five-digit numbers are there so that by deleting one of its digits we get the number 2012?

A) 36 B) 40 C) 45 D) 48 E) None of them

24.

At point A there is a baby rabbit and at point B there is a mother rabbit. The baby rabbit moves 3 units for each jump, while the mother rabbit moves 5 units for each jump. Both of them jump and land at the same time.
According to this, in how many jumps will the baby rabbit and the mother rabbit meet?

A) 11 B) 12 C) 13 D) 14 E) 15

25. The rule of the number sequence whose first 4 terms are given below is as follows:
From the 5th term onwards, each term is the largest digit different from the previous 4 terms.
9, 7, 6, 2, ...
According to this, what is the 2024th term of this sequence?

A) 8 B) 9 C) 7 D) 6 E) 5